

SHAURYA SHARTHAK

NLP Research Scientist

+91 6387864528 | shauryajnvkkr@gmail.com | shaurya-sharthak | IsNoobGrammer | HuggingFace | Google Scholar

PROFESSIONAL SUMMARY

NLP researcher focused on **tokenizer design**, **multilingual LLM adaptation**, and **efficient training infrastructure**. Published first-author research on tokenizer transplantation (**4 citations** in 6 months) with **2× improvement** over baselines. Designed custom tokenizer with **4× better Hindi efficiency** than Phi-4. Founded Tinycompany-AI: **155+ models**, **54 datasets (25M+ rows)**, 3 team members. Built open-source TPU training stack (**234 GitHub stars**). Active research in entropy minimization, RLHF/GRPO, and Indic NLP.

PUBLICATIONS

- Achieving Tokenizer Flexibility in LLMs via Heuristic Adaptation and Supertoken Learning** arXiv:2505.09738 | May 2025
- **Lead Author** (S. Sharthak, V. Pahalwan, A. Kamath, A. Shirawalmath) | **4 citations** | cs.CL, cs.AI
 - Proposed **TokenAdapt**: model-agnostic tokenizer transplantation using hybrid Local+Global embedding heuristic to initialize new token embeddings without full retraining. Achieved **2× improvement** in zero-shot perplexity ratio vs. ReTok and TransTokenizer on Llama-3.2-3B and Qwen2.5-3B.
 - Introduced **SuperTokenizer**: probabilistic multi-word pre-tokenization with Gaussian chunk-length distribution, reducing sequence fragmentation and improving compression efficiency.
 - Cited by “Training-Free Tokenizer Transplantation via OMP” (June 2025) and 3 other papers. Code: 14 stars, 5 forks.

RESEARCH PROJECTS

- QTK-81K Custom Multilingual Tokenizer** | HuggingFace 2025
- Designed custom 81K vocabulary BPE tokenizer achieving **4× better Hindi tokenization** (10K vs. 40K tokens) compared to Phi-4 (100K vocab), outperforming DeepSeek-R1 (128K) by 2.1× and Gemma-2 (256K) by 1.5×.
 - Iterated through 3 tokenizer versions (72K BPE → Bilingual BBPE/SPM → QTK-81K); integrated with **1.83M-row packed math pretraining dataset** for full-cycle TokenAdapt evaluation.
- Indic NLP & Chain-of-Thought Research** 2024 – 2025
- Designed **6 Hindi/Hinglish CoT datasets** (21K+ rows) covering medical, philosophical (Bhagavad Gita), and general reasoning — first structured CoT resources for Indic languages.
 - Created **Be-More-Specific-USER** (51K rows, 9 likes) — novel trilingual clarification-seeking dataset advancing AI query refinement research.
 - Built **Hinglish-Preference-DPO** (9.4K rows, 256 topics) and **Hindi-Instruct-HQ** (28K rows) with synthetic generation via Mistral-Large APIs.
- RLHF/GRPO & Entropy Minimization Research** 2024 – 2025
- Trained **Qwen2.5-GRPO-7B** (471 downloads) using Group Relative Policy Optimization; curated **3.5M+** high-quality math samples for reward-model training.
 - Running independent ablations on Qwen models studying **entropy minimization** during SFT — investigating effects on model steerability vs. surface-level alignment.
- PyTorch/XLA TPU Training Infrastructure** | 234 stars, 27 forks 2024
- Engineered end-to-end training stack for **TPU v2-8** enabling free fine-tuning of **7B–14B** models; built **Optimized-Attention-Torch-XLA** with GSPMD parallelism reducing HBM by 35%.

TECHNICAL SKILLS

Research: Tokenizer Design (BPE/BBPE/SentencePiece), Tokenizer Transplantation, Multilingual NLP, RLHF/GRPO, Entropy Minimization, Pretraining Data Curation, Chain-of-Thought Distillation, RAG

Training Infra: PyTorch/XLA, TPU v2-8, GSPMD Model Parallelism, Flash Attention, Unsloth, LoRA/QLoRA, TRL

Frameworks: PyTorch (Expert), HuggingFace (Expert — Transformers/Datasets/PEFT/TRL/Tokenizers), LangChain

Languages: Python (Expert), SQL, C++, Bash

ORGANIZATION LEADERSHIP

- Founder — Tinycompany-AI / BiBo Inc.** 2025 – Present
- HuggingFace (tinycompany) + GitHub (Tinycompany-AI) 3 team members
- **155+ models** (BiBo-Super-IT, Qwentify3, LQSalad series) and **54 datasets (25M+ rows, ~140 GB)** across medical, math, code, reasoning, and multilingual NLP.
 - Published TokenAdapt and SuperTokenizer code (27 combined stars); active in 4 HuggingFace organizations.

EDUCATION

Chhattisgarh Swami Vivekananda Technical University 2022 – 2026
B.Tech (Hons.) Computer Science (AI/ML) Bhilai, India

Jawahar Navodaya Vidyalaya 2018 – 2021
Senior Secondary (GPA: 3.86/4.0) — Secondary (GPA: 3.6/4.0) Kurukshetra, India